

ECSE-4050/6050 Advanced Electronic Circuits

Course Syllabus

Basic Overview

Course Instructor:

- Kyle Wilt, JEC-6004, wiltk2@rpi.edu
- **Office Hours:** Wednesday 2:00-4:00 PM, Webex or In-Person

Teaching Assistant(s):

- Emre Acarturk, acarte@rpi.edu

Lecture Schedule and Location:

- Monday/Thursday, 10:00 AM - 11:20 AM, JEC4104

Schedule: The class calendar will be accessible from the LMS site.

Overview: Linear and non-linear applications of operational amplifiers, with an emphasis on circuit design. Non-ideal operational amplifier behavior, including both static and dynamic characteristics. Amplifier stability and frequency compensation techniques. Operational amplifier-based oscillators. Circuit noise. *3 credit hours*

Overall Educational Objective: The objective of this course is to study the principles of analog circuit analysis and design with an emphasis on practical applications. The focus is primarily on the use of operational amplifiers and a strong emphasis is placed on the non-ideal aspects of this device and its impact on circuit performance.

Student Learning Outcomes: On completion of this course, the student will: (1) Analyze and design linear op amp circuits (2) Determine the error introduced by non-ideal op-amp characteristics (3) Determine the noise at the output of a circuit containing op-amps (4) Apply frequency compensation to stabilize op-amp circuits (5) Analyze and design non-linear op-amp circuits. Students taking ECSE-6050 will further be expected to be able to: (1) Characterize and compensate for effects of non-ideal circuit components on op-amp circuit behavior (2) Design op-amp circuits capable of driving both capacitive and inductive loads and write a final project report that includes a review of research papers and discussions of their design relative to (in comparison with) the advanced designs in the literature.

Student Assessment Measures: Assessment of student progress in the course objectives will be performed through homeworks, exams, and a final project.

Pre-Requisite Courses:

- ECSE-2050 Introduction to Electronics

Texts:

- Sergio Franco (2015), *Design with Operational Amplifiers and Analog Integrated Circuits* (4th Ed.). McGraw Hill

Online Tools: The following online tools will be used to support this course. If you do not have access to any of these tools, please inform the instructor as soon as possible.

- Cisco Webex: <https://www.webex.com/video-conferencing>
 - Asynchronous communication platform between the students and staff.
 - Used for propagation of course announcements.
- Gradescope: <https://www.gradescope.com>
 - Submission of graded assignments will be done through this platform.
- RPI LMS: <https://lms.rpi.edu/webapps/login/>
 - All course information will be posted here, including the course calendar.

Assignments and Gradeables

Homeworks: There will be ten total homeworks assigned in the course. These homeworks will be a combination of hand calculations and simulations, using spice simulation software (e.g., pSpice or LTspice).

- Homeworks are due as listed in the course calendar. Students are encouraged to work together but must submit their individual work.
- Students registered for ECSE-6050 may be assigned additional homework problems as compared to those assigned to ECSE-4050.

Exams: There are three exams scheduled: two during the semester and one during Finals week. Exams are open-book/open-notes and calculators are need for the exams.

Final Project: The class will culminate in students completing a comprehensive final project. This project is a design contest supported by Analog Devices, Inc. The project will begin early in the semester and students are strongly encouraged to work on the final project throughout the semester. One of more intermediate checkpoints may be assigned as part of the final project. Students registered for ECSE-4050 are allowed to (but not required to) work in pairs to complete the project whereas ECSE-6050 students must work individually.

Course Logistics

Grading: The course will be graded following the table below. There are two grading breakdowns for the course. Student's assigned grades will be the result of whichever grading calculation results in a higher overall grade.

Exams are weighted such that the lowest exam grade is worth 1/2 as much as compared to the other two exams such that the exam average is calculated as $(\text{High} + \text{Mid} + \text{Low}/2)/2.5$.

Component	Quantity	Calculation 1	Calculation 2
		Weight	Weight
Exams	3	70 %	60 %
Homeworks ¹	11	10 %	20 %
Final Project	1	20 %	20 %
Total:		100 %	100 %

¹The lowest homework score will be dropped.

Late Policy: Deadline extensions may be granted under exceptional circumstances and only with prior notification of the instructor or official Dean's excuse. Without an official extension granted:

- Homeworks will be accepted with a 50 % penalty if submitted at any time after the due date.
- Exams must be taken as scheduled.
- The Final Project must be submitted as scheduled.

Attendance and Participation Policy: Attendance is required for lectures; however, no official attendance will be taken. If a class will be missed, the student should notify the instructor to ensure the student does not miss course content.

Re-Grading Policy: All gradeables (excluding the final exam and project) are subject to regrades. Each of these items will be submitted to Gradescope; therefore, all regrades should be requested through Gradescope's regrade request feature. In order to achieve a successful regrade, make sure to be as explicit about your challenge as possible such that the grading error is clear and easily correctable.

Regrade periods for assignments will be limited to 2 weeks after publication of grades. Assignments submitted near the end of the semester will have regrade periods ending after the last study day.

Policy Detail

FERPA Statement: The [Online Tools](#) used in this course provide a service designed to assist schools, teachers and other educational partners to improve student learning outcomes. In some circumstances, these online tools may receive personally identifiable information about students ("Student Data") from the instructor in the course providing this service. For example, an instructor will provide a class roster, email addresses of all students in the class, as well as coursework data that may be linked to a particular student. All listed online resource companies used by the instructor consider Student Data to be strictly confidential and have physical, administrative and technical security protections in place to protect such data. They do not use personally identifiable Student Data for any purpose other than to provide the services to the instructor, and they do not share personally identifiable Student Data with any third party except as authorized or required by the instructor. The online tools above may collect, analyze, and share anonymized or aggregated data or data derived from Student Data for certain purposes, but only if the disclosure of such data could not reasonably identify a specific individual or specific School. Collection and use of Student Data provided by the instructor is governed by Terms of Service for each platform and by the provisions of the Family Educational Rights and Privacy Act (FERPA). Student Data is provided and controlled by the instructor.

Diversity, Inclusivity and Accessibility Statement: Rensselaer Polytechnic Institute strives to make all learning experiences as accessible as possible. We strive to provide a learning environment that is equitable and conducive for all students; regardless of age, race, ethnicity, religion, gender and sexual identities, disability, etc. If any student does not feel their needs are met in this way, they are strongly encouraged to discuss their comments or concerns with the instructor(s) such that we may improve the classroom experience for all students.

Please contact the instructors as soon as possible if you anticipate or experience academic barriers based on a disability such that alternative options may be discussed and determined early. To establish reasonable accommodations, please register with The Office of Disability Services for Students. After registration, make arrangements with the instructors as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion. To receive any academic accommodation, you must be appropriately registered with DSS.

DSS contact information: dss@rpi.edu; 518-276-8197, 4226 Academy Hall.

Collaboration and Academic Dishonesty: Intellectual integrity and credibility are the foundation of all academic work. A violation of Academic Integrity policy is, by definition, considered a flagrant offense to the educational process. It is taken seriously by students, faculty, and Rensselaer and will be addressed in an effective manner. If found responsible for committing academic dishonesty, a student may be subject to one or both types of penalties: an academic (grade) penalty administered by the professor and/or disciplinary action through the Rensselaer judicial process described in [this handbook](#). Three relevant academic integrity violations to emphasize include:

Collaboration: Collaboration is defined as deliberately facilitating an act of academic dishonesty in any way or form; for example, allowing another student to observe an exam paper or allowing another student to "recycle" one's old term paper or using one another's work in a paper or lab report without citing it as another's work.

Copying: Copying is defined as obtaining information pertaining to a graded exercise by deliberately observing the paper of another student; for example, noting which alternative a neighboring student has circled

on a multiple-choice exam.

Plagiarism: Plagiarism is defined as representing the work or words of another as one's own through the omission of acknowledgment or reference. Examples include using sentences verbatim from a published source in a term paper without appropriate referencing, or presenting as one's own the detailed argument of a published source, or presenting as one's own electronically or digitally enhanced graphic representations from any form of media.

The Rensselaer Handbook of Student Rights and Responsibilities defines the full list of forms of Academic Dishonesty and you should make yourself familiar with these. In this class, all assignments that are turned in for a grade must represent the student's own work. In cases where help was received, or teamwork was allowed, a notation on the assignment should indicate your collaboration. If you have any questions concerning this policy before submitting an assignment, please ask for clarification.

Any assignments that have been identified as violating the Academic Integrity policy will be assigned a 0 for all parties involved, with multiple violations or egregious violations resulting in immediate student failure and referral to the Dean of Students.